



Sculptors of Noise: Control in Midjourney AI Art Community

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CS 6470: Design of Online Communities

Abstract

The Midjourney Discord server is an online community for users of the artificial intelligence (AI) image generator Midjourney. With a focus on exploring accelerating new aesthetic possibilities, the Midjourney server has grown at a tremendous rate while facing challenges in onboarding this massive influx of new users, scaling to meet the demands of a more diverse user base, and preventing negative societal impacts. As words become images through the Midjourney interface, distinct communities and styles of usage form in this booming frontier of synthetic image generation. This paper will provide a detailed evaluation of this complex community in terms of Amy Jo Kim's 9 community design principles and support it with evidence from observational and user interview data. Patterns that appeared in the interview with 12 participants, such as banning words and the perceived control of random generation, will be analyzed. Finally, the research team offers design suggestions to the observed major issues including onboarding process and reducing bias in the community. This study presents the rapidly evolving sociotechnical dynamics of a vibrant creative community, who learn to control and make use of a powerful AI art generator via a Discord server interface.

Keywords: AI art, moderation, control, imagination, online community, IP, copyright, Discord, Midjourney

Introduction

"We are sculptors of noise. We sculpt from noise instead of from, say, marble. We use prompts as chisels, with force, angle, strength, specificity, and other ineffable things at play. Our artwork is unusual because so many people look at the chisel and claim that it produced the sculpture, as if the chisel animated itself and tapped away at the noise on its own, without us"

-clarinet

The Midjourney Discord server is an online community for users of the artificial intelligence (AI) art tool Midjourney. Users create their artwork using the Midjourney image-generation AI together in public and private server channels. Midjourney asks users to submit text using the “/imagine” command within specified channels, and the images will be created, such as, “/imagine giant goldfish swimming through forest” (see Figure 1 for the image output) using large GPU server farms spread through the globe. Not every channel within the server allows image generation and are rather designed to support conversation, learning, and fun.

The community started when Midjourney’s Twitter account posted its first tweet on March 21, 2022 announcing that the sign-up for their beta is open and they would start inviting people to join the beta. On April 17, 2022, phase 2 of the beta started, and members of the Midjourney Discord server were able to invite new people to join this community. On July 11, 2022, according to the Midjourney Twitter account, anyone could receive a code to instantly join the Discord server. The number of community members hit 5000 on August 3, 10,000 on August 11, and tripled at the end of August to 30,000. One week later on September 7, when the research team joined the community, there were 1,530,911 members on the server, and the community became the largest server on Discord since then. As of November 4, 2022, the total number of members on the Midjourney Discord server is 3,845,081 (Midjourney, 2022) (See Figure 2). As of the beginning of October, the number of images generated by users across the Midjourney community closely maps onto a power law distribution (See Figure 3).

Related Work

Intellectual Property and AI

The debate of whether AI-generated work can be copyrighted is still undetermined: who would be the owner: the AI, the human who uses the AI, the artist the user is trying to emulate, or the operator of the AI? When Mikalonytė and Kneer explored people's opinions of whether paintings by AI-driven robots are art and whether robots are artists, they used three factors to differentiate cases: the creator, the purpose of the creation, and the object of the creation. They found out that people do not have significant bias towards art created by robots, meaning that "people judge robot paintings and human paintings as art to roughly the same extent". However, according to the authors, though people acknowledge the art done by robots, they still do not consider robots as artists. The authors attribute this to robot anthropomorphism, indicating that people's opinions might change as the robot adopts more human-like features (Mikalonytė and Kneer 2022).

Daniele and Song provide perspectives that are more relevant to the contemporary AI art movement. They emphasized the importance of the device's autonomy during human's use of it in creating art. It is also essential, when analyzing this topic, to make sure that "autonomy in the computational arts should be seen as a spectrum and never as a binary feature." Daniele and Song also raised the issue of creativity; to them, the technical evolution brought on by AI art tools puts artists at risk because they may "confuse the technical achievement with the aim of the artwork, reducing the piece to an empty technical virtuosity." Therefore, creativity and imagination should still be primarily focused when creating art no matter which device people use, and humans should be mindful of the balance of humanity and technology. As Daniele and Song state, "a good work of art is the result of a synergy between the human artist and the tools used" (Daniele and Song 2019).

Lastly, the artists whose works are in the database for training the AI present another facet of the ownership dilemma. This ownership problem could be interpreted somehow from the Shutterstock case when the company started a Contributor Fund to reimburse "artists, photographers, and designers when content they uploaded to Shutterstock is sold by the company to firms like OpenAI in order to develop generative AI models." When talking about the ownership of the AI-generated art, according to a spokesperson for Shutterstock, "because AI content generation models leverage the intellectual property (IP) of many artists and their content, AI-generated content ownership cannot be assigned to an individual and must instead compensate the many artists who were involved in the creation of each new piece of content" (Vincent October 25 2022).

Copyright

Copyright is always a subject that organizations, especially those involved with individual intellectual property, should be cautious of and take seriously. It has also initiated furious debates in the AI art field. The possibly first AI-art-related work that received US copyright registration is *Zarya of the Dawn*, a graphic comic book, effective on September 15, 2022. Its artwork was created by Kris Kashtanova, a New York-based artist, using Midjourney. Kashtanova approached it by claiming that "the artwork was AI-assisted and not created entirely by the AI", and they "wrote the comic book story, created the layout, and made artistic choices to piece the images together" (Edwards September 9, 2022). About one week after this effective registration, Getty Images, an American supplier of stock images and other media works for business and consumers, banned submissions by AI art tools. The concern, according to the interview of their CEO Craig Peters, may be that "selling AI artwork or illustrations could

potentially put Getty Images users at legal risk” and that Getty Images is proactive to protect their users’ rights and benefits (Vincent September 21 2022). Interestingly, another stock media works provider, Shutterstock, which was also reported to be removing AI-generated photos, announced a partnership with OpenAI, the owner of DALL-E another AI art tool on October 25, 2022. Shutterstock will accept and only accept submissions generated by DALL-E because for those art by third parties, “it can’t validate the model used to create the content so can’t be sure who owns the copyright” (Vincent October 25, 2022).

People are also speculating about the possibility based on available relevant copyright cases. According to Ryan Meyer, the Counsel in Dorsey’s Intellectual Property Litigation Group, in a recent editorial, “copyright law does not protect AI-generated artwork, so neither you nor the AI company has any rights in the image” in the U.S. (Conway September 28, 2022). A board from the US Copyright Office also explained their consideration when reviewing AI art in a response to a recently rejected request to copyright AI work: “the courts have been consistent in finding that non-human expression is ineligible for copyright protection” (Robertson February 21, 2022). It seems that the board emphasized the involvement of humans in the creation of a work, meaning that human-leading with AI-assisting work could still be copyrighted; this may explain the successful copyright registration of *Zarya of the Dawn*. The current legal uncertainty leaves AI art and their creators in an uneasy position, but the successful case and the requirement of human authorship may assure people that there is still some degree of protection while everyone is waiting for the law to catch up with this fast growing technology.

Community Moderation

In this paper, we will discuss how the Midjourney moderator team handles content moderation. There have been numerous studies in the content moderation space, with some works also incorporating analysis of online harassment and free speech online. Jhaver et al. state within *Does Transparency in Moderation Really Matter?: User Behavior After Content Removal Explanations on Reddit* that, “Unfortunately, the black-box nature of content moderation on most platforms means that few good data are available about how these platforms make moderation decisions. This makes it difficult for end users, particularly those with low technical expertise, to form an accurate mental model of how online content is curated” (Jhaver et al. 1-2).” This paper goes on to discuss the importance of explanations in the community moderation space, specifically to teach new users what behavior is acceptable and expected. Their arguments reiterate the need for deeper understanding of community moderation practices and standards. Our paper intends to similarly “fill this gap by examining the concepts of transparency in moderation through a large-scale analysis of Reddit data” (Jhaver et al. 4). Notably, this paper found that users were more likely to abide by community behavioral rules when they receive explanations why their content was removed (Jhaver et al. 19). Similarly, Roberts believed, “When those in-house processes are kept from users, as they almost always are, the logic for the existence of the material on a site becomes opaque and the content becomes normalized” (Roberts 3). Our exposure to the auto-moderation tools the Midjourney moderation team uses was limited; nonetheless, there have been multiple studies that cover these types of tools. Jhaver et al. explain in *Human-Machine Collaboration for Content Regulation: The Case of Reddit Automoderation* that bots and automoderation practices help spare moderators from the emotional toll their work brings (Jhaver et al. . Bryan

Dosono and Bryan Semaan studied *Moderation Practices as Emotional Labor in Sustaining Online Communities: The Case of AAPI Identity Work on Reddit*, and they found that “Moreover, since moderators are subject to rote and monotonous tasks in their moderation workflow, they are constantly exposed to disturbing content that may have long-term effects on their mental health” (Dosono and Semaan 9).

Research Questions

The team developed their research questions separately, and then assembled a final list of X questions. These questions cover a range of topics that the team was jointly interested in: the community experience and education, the Midjourney AI algorithm, prompt craft, ownership of AI art, banned words, community moderation, user demographics and motivation:

1. How do you get the AI to make the image you're imagining?
2. How do users explore new possibilities for creativity, art, and design through prompt craft? What community is formed around this exploration?
3. What do Midjourney users think about art appreciation and valuation when it's generated by AI compared to a human?
4. What are people's intentions in using Midjourney? Do users realize how impactful this tool could be to the future of AI art in society?
5. How does the moderating team combat the toxicity in the community?
6. What motivates an individual volunteer to contribute to the community?
7. How do the members and the moderating team (including the founder) perceive each other?

Methods

The research team actively participated in image generation and discussion on the Midjourney Discord Server. The team spent well over 100 hours participating in generating images, observing other users on Midjourney, sharing images, and informally discussing topics of interest with community members. Participant observation served as a foundation for recruiting participants for interviews.

Some of our interviewees requested to use their Discord username in this paper while others wanted to be identified with pseudonyms or their real names. The team interviewed 12 users in total, with 3 moderators, 1 guide, 1 regular user, and 5 super users. All participants are over the age of 18. The total length of interviews conducted is over 15 hours. The research team developed a list of interview questions based on topics including the general experience of the community, moderation, ownership and copyright, banned words, motivation, user backgrounds, and more. The focus of the interview guide was adjusted depending on the research questions, as our knowledge and familiarity with the studied community increased. The interview guide also adjusted depending on the user roles of the interviewee, their distinct style of usage, and their professional background. Based on the interview guide, the team conducted semi-structured interviews via both audio call and synchronous text chat, with the length of the interview never exceeding 90 minutes. The interview is recorded with the consent of the participants, and the team removed confidential information. After the interviews, the team transcribed the audio and analyzed the information. An iterative thematic analysis was applied across interview results which directed a bottom-up analysis of key concepts. The team then applied their findings from interviews and background research to Amy Jo Kim's 9

Principles of Community Design. Applying information to the framework enabled the team to draw design recommendations for the Midjourney community-at-large.

Cody Boy

Cody Boy began using Midjourney in February 2022, after about a year of trying out different AI text-to-image generators. David Holz personally invited him to the Midjourney server, which at the time had just “a couple thousand” people. He is now a moderator. While he is a part of the 10,000 club, he recognizes that he now spends most of his time in the community moderating rather than generating new art. He has dedicated much of his professional career to creating technical tools for artists in the gaming industry, though he currently works in the AI field unrelated to art. He tends to hoard his prompts, and he does not showcase his own images too often. He stated, “I tend to find something that I really enjoy and then just beat it into the ground and make it 1000 pictures out of it.” He believes the moderating work he does is important to the community and the Midjourney team, and he greatly enjoys the moderator community.

Fractl

Fractl began using Midjourney in May 2022. He took a month off from work to give himself time to learn and delve into the world of AI art, specifically to illustrate short stories he generated with GPT 3. He networked with people ingrained in the AI community which resulted in an invitation to the Midjourney community. He has held his own art shows locally showcasing his work which included a walkthrough of how to use Midjourney. He was drawn to helping others in the community, too, which led him to become a moderator. He develops training

materials for his company, and he does the same for the Midjourney server. He still sets aside time to generate art, and he is in the 5000 club. As a moderator, he spends a good portion of his time reviewing user reports and moderating content; he stated that the moderator team generally bans about 5 users every day, with many more users earning timeouts daily.

Spruder

Spruder joined the Midjourney server in July 2022, and distinctly remembers waiting a week to watch others' usage of the tool before he began using his free credits. When he finally felt informed enough, he burnt through his free credits, and then the \$10 plan credits in his first day. He shortly thereafter bought the #30 unlimited plan - "I was immediately hooked." He now is in the 25,000 club, and he maintains an online store selling posters and apparel of his Midjourney creations. He's always loved art and creating, and Midjourney lets him quickly create art whenever he has spare time. He has a background in computer science, and now works around the music industry. He noted that he almost exclusively uses Midjourney through the Discord mobile app. He maintains a few private channels for specific projects or moods, and he typically avoids the general channels given the volume of content created in there. He does not have any formal roles in the server, but he has learned a great deal about crafting prompts and is happy to share his lessons learned. He spends his time in the 25,000 club and in private servers with others he's met in the super user clubs.

MariusJuston

MariusJuston began using Midjourney in July 2022. He has always been interested and involved in AI and machine learning, both educationally and professionally. The research team

found him because he created and still maintains a massive artist style guide, “Open Library of Artist Styles by MJ Community”. This spreadsheet contains trackers of progress made training Midjourney on over 2000 different artists’ styles. MariusJuston developed the artist style guide in part to understand what styles are possible and to teach how machine learning works to those who clearly do not understand. He created the artist style guide without any intentions of getting a formal role or recognition for his work. He is a member of the 10,000 club, and he spends most of his free time in the super user clubs. He generally uses the Midjourney bot in a direct message; he cited his reserved nature as a reason why he more often views conversations instead of contributing, both in showcasing his art and partaking in conversation.

Fnuckle

Fnuckle joined the Midjourney server on June 14th, 2022. Fnuckle works as a full-time motion designer and also creates a lot of experimental art. They heard of Midjourney from a good friend who is also an artist. They asked for invitations on their Instagram and joined when there were around 40,000 members on the server, and each member could only send 5 invitations in one month. One day, Fractl stopped someone’s off-topic monologue in a channel, and a day after that they got invited by a developer they knew to become a moderator. They recalled that in the beginning, community members were already in the creative industry, and “there were a lot of really impressive names in there”. Fractl felt that members were “a lot nicer and a lot more grateful to be there” because it was hard to join the community and people did not want to get kicked out. As one of the head moderators, they oversee the channels they are responsible for, help test new features and banned words, and discuss

moderation issues with the moderator teams. They also run the Facebook group and the subreddit of r/Midjourney.

Clarinet

Clarinet joined Midjourney in June 2022. He treated Midjourney as a place to distract himself from grief due to a life event, a place that “helps you express”. He was using Night Cafe beforehand, and he found Midjourney has an algorithm that works better for his prompts and aesthetics, so he became obsessed with Midjourney. He then, as described himself, happened to be knowing more than other new members due to joining earlier, so he became a “teacher” just by accident. Clarinet considered himself already being a guide before the Midjourney team made him become a guide officially. He is also a part of the 25,000 club. He is actively helping people with prompting-related questions and organizes most of the posts in the “prompt-craft” forum.

Hematita

Hematita joined Midjourney in September 2022. She is a fulltime freelancer specialized in graphic design. She is a huge fan of the movie “Labyrinth” and saw someone posting artwork of “Labyrinth” generated by Midjourney, so she joined the server and started the trial. She then upgraded and went through her allotment quickly. She is in the 25,000 club. She hangs out mostly in the channel “prompt-chat” where she prefers over the forum “prompt-craft”. Hematita is using Midjourney to make money. Hematita, as other regular members, calls Midjourney “Midge”. To her, it is “for the personification. And thus calling Midjourney "her" and "she". It’s fun to personify software like this, especially when it's part of a creative

process.” Hematita also owns her own Midjourney server “Midjourney Madness”, with the all-women moderation team, for her to share interesting images and have slower discussions with a smaller group of people invited by her, avoiding the chaos of the main server. She makes sure to check people’s pieces to avoid those who chase nudes or porn. She is actively seeking women and her goal is to have her server be at least half women.

Martina

Martina is a UX designer from Spain who has worked in design agencies and the web3 space. Currently, she runs a YouTube channel and social media account that profiles new developments in XR and AI technology for design. Her posts on social media introduced one of the researchers to Midjourney and AI art over the summer of 2022. Martina was active in the Midjourney community in June and shared her knowledge of prompt craft through projects and guides on a variety of other social media platforms. Recently, she has taken a break from AI art, but plays an important role in the Midjourney community ecosystem as a “design influencer” who has widely publicized the platform and developed tools to help new users get started.

Joseph

Joseph is a creative director for an indie video game studio. The studio has had success in the indie game space, and Joseph is currently leading the development of an innovative VR game. Joseph reached out to one the researchers after we asked a question during office hours about how best to cede creative control to the AI to encourage original image generation. Joseph brought up an experience he had discovering a “magical prompt” that had helped to solve a design problem in the VR game. This “magical prompt” created an innovative aesthetic

and architectural style that he explored through the generation of over 10,000 images with Midjourney. As a super user with over 25,000 images generated in Midjourney, Joseph helped connect the research team to other super users he considered to be regulars in the 25,000 club. Joseph is distinct from other users interviewed in that he keeps the majority of his prompts secret to maintain a period of exclusivity on them before the forthcoming VR game is announced. His usage of Midjourney was distinct from other super users interviewed in that it was “99% professional”.

Ignite

Ignite is a super user on Midjourney who has made over 25,000 images. Ignite works in health technology and is an active member of the Discord community who likes using Midjourney throughout his day. He has moved through several styles of usage from more targeted and specific to more exploratory and random. With a background in technology development, he has developed a series of protocols for random image generation which live in an Excel sheet full of hundreds of different artists, styles, and nouns. His constant experimentation has allowed him to develop a unique perspective on Midjourney, the possibilities inherent on the platform, and the challenges this technology holds for society.

Oscar

Oscar was called the “David Bowie” of Midjourney by Joseph who introduced him by explaining he was, “extremely talented and does really silly funny things, and if it turned out to be that that person was actually an alien, it would make perfect sense.” His background in technology gives him insight into the inner workings of the algorithm, that allows him to

explore the “latent space” of Midjourney. Much of his experimentation works through and around Muppets being placed in curious places. He collaborates with a group of users who builds worlds around a character called “yellow Elmo” or “Yelmo” that gets dropped into a multitude of different creative situations and stories (See Figure 12). He has done several different tests to determine different stereotypes and biases in the algorithm. During the day, he works in health technology.

Shambibble

Shambibble is a patent attorney outside of being a prolific creator on Midjourney. His experience describing the technical dimensions of intellectual property and “thinking adversarially” has allowed him to make several breakthroughs in prompt craft, most notably, recurring face image prompting and word reinforcement. He is part of the 25,000 club and has been writing a book on prompt craft for Version 3 of Midjourney. He is an active tinkerer and enjoys helping other users sort through technical prompts which require carefully controlled parameters. He is rigorous and empirical when it comes to prompt craft and offers patient and humorous explanations of the inner workings of the AI.

Community Evaluation

The research team evaluated the Midjourney Discord server using Amy Jo Kim’s 9 design principles of community design (Kim 2000), which are the titles of each of the following sections:

1. Does it have a clear *purpose*?

Audience

David Holz, the CEO of Midjourney, stated the average age of the members currently is 33 with people under 18 and also above 65. Members join this community mostly for the same primary reason: to create images by submitting text to the Midjourney AI tool. However, the purposes of image creation are diverse. According to David, about 30% of users are professionals, such as graphic artists who use Midjourney to create concepts and seek inspiration (Claburn August 1, 2022). This was back in August 2022. The professional users in Midjourney have shown more potential for this tool. For example, the cover of *The Economist* in June was created by Midjourney (2022). Someone submitted a piece of work generated by Midjourney and won first place at a state fair fine arts competition (Gault August 31, 2022). There is also a group of people who may not have adequate or professional art skills to create art by themselves, so they turn to Midjourney and leverage its power to “visualize” their imagination. Members in this community are also here for fun, creating images just to explore how powerful the tool is, for therapy to generate images inspired by deceased loved ones, from humans to companion animals, to entertain friends and children, etc. Most interviewees restated David’s stance that Midjourney is “for everyone.” There are a number of sub-groups that users self-assign to, which will be discussed later in the analysis.

Mission

On the home page of the Midjourney Discord server, it says: “The official server for Midjourney, a text-to-image AI where your imagination is the only limit.” This server’s primary mission is to provide a space for users to create images from the text they submitted, exploring

artistic expression of their imagination and creativity. Besides image creation, this Discord server also provides channels for people to seek help with creating images, discuss Midjourney-related topics, and showcase their works of art. The flow of content in any channel makes this purpose clear, as generally, channels are flooded with images, and conversations tend to only focus on Midjourney and AI art. The community-at-large could be considered a cradle for creative works that are considered art and to make humans more artistic; for example, it is common to use Midjourney to illustrate personal short stories, comics, and more. When asked about his attitude towards the opinion that AI art tools are replacing human artists during one of the office hours events, David responded that he does not want to make an AI that replaces people, but one that makes humans better. While the mission of Midjourney is easily discoverable, it is entirely possible that not all users are aware of the mission due to the possibility of someone beginning to use Midjourney before reading through any guides or rules that are placed around the Discord server. Additionally, the mission statement is slightly misleading, as there are limits on what kinds of content can be generated, which will be expanded upon later in the analysis.

Visual Design

Though the design and the interface of this community are largely aligned with Discord's standard appearance for a server, Midjourney still encourages the members to explore their imagination in many ways. For example, the icon of Midjourney is a sailboat on the ocean, meant to illustrate that users can imagine a journey to take them wherever they would like (See Figure 4). This is also reinforced by the command members use to tell the Midjourney bot to start creating the image. Compared to the "generate" button in DALL-E, another AI art tool,

“/imagine” is the command users type into the chat box. Super users have found that there seems to be a common color palette that is applied to Midjourney imagery when users do not specify their prompts with specific colors or branding (See Figure 5).

Backstory

There are numerous AI art tools and communities, and Midjourney was not the first of its kind. The backstory of Midjourney is not officially posted anywhere. The research team was unable to find a thorough timeline of Midjourney in one place, rather we had to stitch together stories and tidbits from different conversations. A timeline of version releases is included within the analysis about banned words.

One noteworthy fact about Midjourney which is not frequently recognized by the members and the public, is that Midjourney is still in its beta phase. However, because of its public accessibility, its successful high-resolution imagery, and its subscription plan, current members in the community, including the research team, join the community and treat the program as if it is mature, well-researched, and well-established product. Clarinet shared,

“... there's two clearly non-product parameters, “-test” and “-testp”. People are constantly like how come I can't do this with test and I'm like, OK, what just came out of your mouth? “I can't do it with test.” I'm like, uh-huh. It's a test. It's literally a stripped-down version of a particular little clump of things that they're testing. We're testing it.”

Still being a beta version means premature design in multiple places in the community, especially when the team is mainly focusing on improving the algorithm. It also brings more

frustration after overestimation and expectation when members are not clearly made aware of this fact. Clarinet also pointed out the shortage on resources that leaves problems unsolved,

“...the reason why you don't know and you're not told is that it's just a beta. And if we took the time to invest in really nice onboarding, it wouldn't make any sense because it's not a product yet. It's changing literally every week [and] something new happens, but it doesn't scale as a business. We would be spending all our time tweaking and fixing ... A lot of the problems would be solved by education, but it is premature to set up an education service arm.”

2. Does it have distinct, extensible *gathering places*?

Places, Features, and Galleries

On a Discord server, there are usually multiple channels created and categorized by the moderation team in terms of purposes. There are also some sections that are not visible to everyone and require certain criteria for members to be included. The 10 public sections are for different purposes:

Info: This is where the Midjourney team makes announcements or sends out surveys for any member feedback. Members’ ability to send messages in the channels in this section is disabled.

Image gen: There are “newcomer room” channels that are accessible to all users, which are meant for trial users to generate images within. There are also 20 “general image gen” channels

for all paid users. These channels typically feature no discussion, and only an onslaught of images generated by other users.

Support: When members encounter problems, mainly technical problems or subscription questions, they can receive help in these channels.

Chat: The section contains mainly discussion. Members give their feedback to the moderation team in the “feedback” channel, discuss Midjourney-related topics in the “discussion” channel, get into debates or even arguments on questions like “should AI-generated images be considered art” in the “philosophy” channel and chat more freely about anything in the “off-topic” channel. You are not able to generate images in any of these channels.

Community Forums: Discord released a new feature in 2022 that creates and collates multiple forums. The forums provide a more exclusive space for people to review and offer information so those pieces of the information are more readily visible for longer and can be easily searched instead of being flooded by new messages. There is the “prompt-craft” forum which is the main space to request and offer help on creating prompts and testing features and hacks. The other forum is “image-jams”, which features themed forum threads that people can post their relevant creations within.

Showcase and Image Galleries: Images are showcased and admired here. Members post their images or their re-creations, including but not limited to digital and real-life work, in different channels under the section “showcase”, and send the images they like to the “favorites” channel under the section “image galleries”.

Themed image gen: This is another public section for members to generate images that fit different themes or topics, such as ‘dragons’ or ‘anime’.

Thread Channels: Members can also create their own threads under different sections and “thread channels” is where members could view records for thread creations with the ID of the members who create the thread.

Voice Channels: This is a default feature of a Discord server where members could do voice chatting. This is also the stage for the weekly office hours, which will be discussed in detail later. While there are <10 voice channels in existence, they do not typically have any more than 5-10 users within them at a time. Based on our team’s observations, they are noticeably less used than most other channels, which is why our team did not observe these channels frequently.

There are also some sections that are only accessible to members who meet certain criteria, such as the super-user-clubs. There is also a set of channels that is restricted to moderators and guides, which the team received limited knowledge on. Moderators have access to a few channels with automoderation bots that create logs of reported images and users. There is also a “tavern” channel for moderators to socialize, and discussion channels to discuss features they are testing or to discuss observations about the community. The moderator channels will be discussed further in a later section.

Super-User-Clubs: The section “super-user-clubs” is only open to members who create a certain number of images. Currently the five channels are “1000-club”, “2500-club”, “5000-club”, “10000-club,” “25000-club.” As more exclusive spaces that require a certain amount of

dedication to unlock, these clubs act as platform elements that make the platform more of a game. People report “gaming” the algorithm by running `--q .25` prompts to increase their number of prompts (`--q` values determine how long the AI spends on the image with `.25` being the fastest, least-detailed setting, and `5` being the slowest most detailed setting). Part of the push to get into the exclusive spaces is a sense that “arcane knowledge” is being shared there and that being part of a group of seasoned users will improve one’s prompt craft. Users like MariusJuston and Spruder both cited the super-user-club channels as the places where they have learned the most, and now spend the most time. As these spaces become more popular users reported that they became “gentrified” and pushed the more artistic super users to seek out even higher-level clubs to maintain the conversation discourse amidst ever growing “walls of images”. The “gentrification” of these channels will likely lend itself to another new super-user-club channel, which will provide a haven for true Midjourney fanatics to socialize and share knowledge, even if for a limited time (Oscar).

3. Does it have *profiles* that evolve over time?

Currently, the beta version of Midjourney has been open to everyone since August. Anyone who has access to the internet will be able to join the Discord server with a Discord account.

Entry

Everyone will start with 25 free images. After the members use all their free trials, they will be offered plans for subscriptions to continue to generate. The members can find the information by clicking on the link in the “faq” channel and going to an external page where

relevant information is well-organized and displayed. The information could also be sent to the members when they are asking for more than their current membership provides, for example, more images when using up their free trials. Or users could use commands like “/info” and “/settings” to view the information. The message regarding membership and subscription will be sent to the members by the server bot in the current channel that users submit the command. Excited users will likely feel limited by 25 images, which could push them towards a subscription quickly. Not much personal information is collected in order to subscribe to the Midjourney server beyond payment information, and the personal information one already provides to Discord.

History

Discord makes it very easy to search all communication a user has sent within the server. Additionally, user’s galleries will show all images the user has ever generated, excluding any images that have been hidden by moderators. The images are automatically added to the user’s gallery upon generating. There are 4 sections on the member profile. The header section displays the member’s name, Discord ID, and profile banner. Members could also write an introduction of themselves in the “About me” section. The date they joined Discord and the Midjourney server are shown right underneath. The last section is “Roles”, where there are several tags indicating the members’ groups, roles, and identities. Many users include links to personal galleries or shops to display their Midjourney-generated art (See Figure 6).

4. Does it promote effective *leadership*?

The daily running of the Midjourney Discord server depends largely on the volunteers, which are the moderators and the guides. Though there are moderators who are appointed by the Midjourney team, most moderators and guides have their own jobs and are volunteers. Besides moderators and guides who may provide technical support for the members, developers from the team role sometimes also come by and answer technical questions from members. Super users also commonly provide knowledge to others. There is no official mentoring program, nor an official process people can follow to become a moderator or guide.

As fnuckle described, being a moderator is to “be the bad guy to an extent”. There is no standard or formalized way of moderation. Each moderator has their own style of approaching people. For example, some have a master list of messages that they send to members who violate the rules ,and some moderators spend time communicating with the upset members to calm them down from negative emotions due to being warned.

Moderators are also responsible for taking care of each other's behaviors when some moderators are approaching members in an inappropriate way. For example, fnuckle mentioned a time when one moderator said to a member that if they ask questions that are not included in the survey again, they will be blacklisted. And fnuckle immediately deleted that message because, as they said, “don't threaten people to put them on a blacklist.”

5. Does it have a clear, yet flexible *Code of Conduct*?

Midjourney has three different pages that list members’ policy, rights, and their terms of service. The Midjourney Discord server has a channel called “rules” to display its terms of service. On the external official website, Midjourney provides a detailed [privacy policy](#) to tell

members what data are collected from them and how Midjourney uses the data, as well as a terms of service page. The terms of service mainly address problems of information and data, copyright, and community behaviors (“Privacy Policy”, “Terms of Service” 2022).

Information and Data

Midjourney makes it clear that members should be at least 13 years old and in the age of digital consent in their country. They have admitted that Midjourney tries to keep a PG-13 and “family friendly” environment, but they cannot guarantee and control the assets created by other members. The other information collected by Midjourney are email addresses, usernames, billing information, image outputs, favorites, and text prompts or images that members submit to the bot for image generation.

Copyright

In the Midjourney Discord server community, members, including some of our participants, profit from the images they created using Midjourney. In their terms of service, Midjourney claimed that paid members “own all assets” they create with the services. It is also clearly noted and widely, though not always happily, accepted by the members that the images and prompts are publicly viewable, reusable, and remixable by other members, unless created in a private plan. For non-paid members and corporate-user members, there are separate terms that address privacy and profit issues. Midjourney primarily uses the private mode to help address privacy and copyright issues. According to our interview, Midjourney has not yet had a great solution to address copyright violation between members. Clarinet talked about his art being stolen,

“There are people who go and scrape all of the images off of Midjourney’s gallery. They pay for membership for the privilege to do this because you can't see that gallery without paid membership and then they pull down all of the high res images and then they open a shop. My stuff's been pulled off of NFT shops all over the Internet. They did not learn to prompt. They did not create anything. They just opened an Etsy shop of stickers, and they made stickers of what you made and didn't ask you.”

It is not Midjourney’s fault that they do not have a quick response because, as Clarinet also said, the AI art is still a new thing, and “nothing gets resolved in that amount of time. But not because they're stupid or bad or wrong, it's just because the world doesn't move that fast.” Furthermore, the responsibility of protecting members’ copyright and ownership of their works does not fall on only Midjourney or other AI tool owners; platforms like Etsy in this case should also be taking care of the submissions members provide to avoid any violation to others’ properties. Cody Boy described prompts as “trade secrets”, and users should decide if they want to keep their own or not. Discord encourages users to share their prompts to make Midjourney a social experience.

Community Behaviors

The guidelines for community only call out 5 rules in the Terms of Service. For example, the guidelines ask members to not make adult content or gore followed by a short sentence mentioning that some text inputs are blocked. Unfortunately, the continuous creation of (subjectively) inappropriate images and conflicting perspectives around banned words is causing a great amount of discussion and debate. This complicated topic and its associated

dilemmas will be expanded upon in the later section. Besides content-related behaviors, the guidelines also mentioned respectful interaction between members, and no violence or harassment is tolerated. Again, this rule seems to be too vague and broad to address the diverse ways of interaction between members. Hematita once tried to start a serious conversation about unsolicited direct messages with pornographic pictures. Her requests for help were belittled. Even though a moderator reached out to her, she seemed to be disappointed by the other members' off-topic response in the feedback channel.

6. Does it organize and promote *cyclic events*?

There are many different forms of cyclic events in the community. Most of them are organized by the Midjourney team.

Weekly Office Hours

The office hours is probably one of the events that has the highest participation and associated expectations. It starts at 3:00 PM EST via the voice channel on Discord. Community members receive a reminder on the event page every week (See Figure 8).

The normal office hour starts with David providing updates on the current work the team is doing, any noticed challenges or problems, and upcoming, exciting new features and releases. The "Q&A" section follows; before and during office hours, a survey will be sent out to collect people's questions for David. Moderators will help add members who want to get on the "stage" to verbally speak to David in a queue and bring them on the "stage" one by one during the "Q&A" time. For those who have questions to ask but do not want or to or cannot be on the "stage", a moderator will also join the queue to ask questions for them. In the meantime, the

audiences (both members and moderators) could chat in the “vc-text” channel and David will also check in there when he speaks.

In a survey description, the team says: “new/interesting/fun questions will be prioritized”. Most members ask very technical questions, for example, “How is the GPU used for image creation?” Members are also curious about philosophical questions like “How to define what beauty is?” and “What is creativity and who is creative?” They also give suggestions to David showing their desires for a certain feature like a web API and a place to show appreciation to others. As members get more familiar with Midjourney, the community, the flow and vibe of the office hours and even David, there are some new types of questions. Members have asked personal questions, from very basic, “Are you a cat or dog person?”, to somehow in-depth, “What helped form you today?”, which led David to share a long story of how his life of walking along a beach at 15 in Florida makes him think about who he wanted to become and become the person he is today. What is interesting is that, when members apologized for asking questions that are interesting but seem to be off-topic, others would respond by saying it doesn’t matter because David likes that type of question or they themselves also like the questions. This helps solidify a virtuous leader of the community who makes himself somewhat accessible to all 3 million users.

The whole office hour usually lasts about 4 hours with around 1500 people joining (this number is recorded for those office hours we joined). Members who spoke during office hours are from countries including the United Kingdom, Sweden, Germany, etc. There are also members actively taking notes of the office hours and sending them to the channel for those who missed or could not make it to the office hours. When asked about recording the office

hours, David himself is reluctant because he does not like people taking his words and interpreting him out of context.

Daily Theme

Besides the office hours, the daily theme also draws many participants. It is a very straightforward event, where every day a moderator will announce a new theme in the channel “daily-theme” under the section “themed image gen”. Members create images of that theme. The previous themes include “cat”, “night”, “--seed 1026”, etc. The Midjourney team provides instructions to let the members opt out of the pings in a pinned message in the “daily-theme” channel as many members seem to be annoyed by the daily notification.

Micro-polls

In the channel “micro-polls”, David posts quick questions as surveys to ask for people’s feedback and opinions. The questions are mainly technology or service-related, for example, people’s preference for the plan, people’s needs for a certain new feature or mode, etc. There are also community organization-related questions like whether the team should add a new channel and how people feel about the activity levels in the members' chat rooms, which the majority of people feel are overwhelming. Sometimes David will also ask interesting questions related to Midjourney the product, like what style people perceive as the Midjourney style. We have not yet seen evidence of how the polls influence the Midjourney team’s decisions on building the community, but the results of the polls were mentioned during the office hours.

7. Does it provide a *range of roles* with increasing involvement?

The roles in the Midjourney Discord community are diverse. There are 5 main categories based on the Discord “Member list” panel: Team, Bot, Moderator, Guide, and Users.

Team: The team role refers to the groups of people who own Midjourney and develop Midjourney, including the CEO David Holz, advisors, and developers, whose names could also be found on the “about” page on the Midjourney official website. They have exclusive power over the AI tool. They also do not have to pay the subscription and are paid by Midjourney as employees. They have unrestricted access within the community. Most join the team by specifically applying to a job posting on the official website. Previously, at least one moderator joined the team after expressing interest.

Bots: There are two public *bots* in the community. The “Midjourney Bot” is responding to members’ commands with the corresponding actions, for example, generating images when received “/imagine” and displaying account information when received “/info”. The “Charon the FAQ Bot” mainly automatically answers members’ questions by detecting keywords in messages they send in the channel. There are additional bots exclusive to the team, moderator, and guide community that are explained later within the banned word analysis.

Moderators: They are responsible for regulating activities and discussions in channels and solving problems in running the community. Their work includes but is not limited to sending warnings to members with actions that violate the rules, keeping the discussion in the “chat” section on the topic, and responding to members’ feedback regarding the community. Moderators have internal communication channels for them to discuss new rules, features, and any topics related to community regulation.

Guides: There is limited difference between guides and common users except that they are voluntarily helping others. They are usually people who have more knowledge on certain aspects of Midjourney or the community, for example, prompt crafting, and who are willing to answer others' topics and keep the content organized. Guides tend to be more focused on helping users achieve what they are hoping to accomplish, whereas moderators may be more directly involved with the team and the tool. Moderators and guides are not rigidly distinguished from each other.

According to our interview, the *team* invited early members who were proactively helping or moderating the community to become moderators and guides. Then the early moderator team looked for members who could be promoted to become a guide or a moderator. Moderators and guides will typically flag any user within the general community that has the potential to take on a role, and it takes 3 moderators to approve appointing a user as a guide. Historically, most moderators and guides do not actively campaign or ask to become their role. Also, the moderators look for patient, calm personality types, given the high volume of repeated questions these roles receive from other users.

Among typical members who do not have an "official" role, there are also different groups in terms of their knowledge and usage of Midjourney and involvement in this community.

Visitor: People without a Discord account who just visit the official page of Midjourney, they do not have access to this Discord server. However, they are able to view the community's

showcase on the external Midjourney official website, where the recently popular and top-rated images created in the community are displayed with their prompts.

Newcomers: “Newbies” are people who join the server with their Discord account and do not pay for a membership. As mentioned previously, newbies will have 25 free image generations and access to only the “newcomer’s rooms” for image generation and “trial support” for getting help.

Member: People become members when they pay for a plan and each plan has different resources and access.

Two main types of subscription are \$10/month for 200 images, and \$30/month for unlimited (See Figure 7). There is also an enterprise membership that charges \$600 annually. The number of image creations and the amount of GPU time would be the major differences and benefits of memberships.

Super User: Among members, there is a group of people who are addressed as *super users*. Those people are members who created at least 1,000 images. They have access to the section “super-user-clubs”. The moderator team previously considered creating a role to denote super users, but ended up never submitting the idea to the team after their inability to agree. They could not agree on the right balance of acknowledging people’s hard work and improperly glorifying people who likely could do more for the community. As Fractl described it, “It was hard to find a way to make it an officially recognized but not unofficially sanctioned role.”

8. Does it support member-created *subgroups*?

Threads

Besides the official channels set up by the Midjourney team, members are always free to create their own threads in each channel. The threads normally have less visibility compared to the channels. They are folded into a page which will appear after someone clicks on the threads icon next to the name of the channel on the top of the Discord window. Threads in each channel are different, and Discord prioritizes the recently active threads by showing them first. Active threads are mostly created in the channels related to image generation in the “themed image gen” and “general image gen” sections. It seems that members are using threads as their own space to create images so their work does not get lost in the huge wave of others’ images. There are also members creating threads to play with image generation with their friends.

Special Subgroups

Members get to know each other and end up forming implicit subgroups in places they can socialize within. For example, in our interview, fnuckle mentioned that they have noticed there are a core and regular group of people usually around in the “prompt-chat” channel and they are positive and helpful. The moderator team had some issues with certain subgroup in the voice chat because they are “using the voice chat just as their personal like streaming platform”.

There are also subgroups outside of this Discord server. During our interview, our participant Hematita shared that she made her own server about Midjourney and she handpicked people to invite them to it. They usually share their images, chat, and give each

other ideas and help on that server. She started this to have a less busy place with a smaller group with a goal “to let it get to a point where we don't often have silence, but also don't feel overwhelmed by the rush of posts pushing everything away quickly.”

Not everyone seems to be optimistic about having many subgroups in the community. In the interview, Clarinet expressed his concern about the setup of Community Forums since it seems to lead to more isolated and unattended discussions. He also proposed potential design suggestions but has not heard back from the team,

“Each of the forum threads is also treated like a chat space and so our users who used to all be together in a community, in prompt craft the chat are now distributed among 134 empty isolated unattended chat channels in the Prompt craft forum. So it destroyed the community and that's a strong way of putting it ... And here's how to fix it. Rename the prompt FAQ forums to FAQ. Call it the prompt FAQ. Make it read only so it's searchable, discoverable and linkable. You can have all your FAQs, but have all of the community chats focused in a chat area so that everyone is brought together again in a crowd. And it's not just the allocated one or two people and empty unattended silence.”

9. Does it integrate what's online with the *real world*?

Though we have not collected much information about members connecting in the real world, according to one of the office hours, the Midjourney team flew from places to gather together and have barbeque. David also mentioned the fact that many team members actually have not met with others and did not even know what each other looks like. They were asking usernames to recognize each other. And the team may have more local specific city events.

Ultimately, since this community is not inherently social, personal connections are infrequent looking at the community at large. While the community would likely celebrate anyone who shared personal achievements or holidays, the non-Midjourney-related conversations can be limited. If anything, features like the daily theme or threads can connect others with real connections, be that through celebrating holidays or identifying and discussing shared interests. Interestingly, every office hours announcements tend to feature a number of emoji reacts from users that pertain to identity, such as country flags, pride flags, and religious symbols.

Continued Analysis of Observational and Interview Data Trends

‘The Banned Words’

Online communities in the United States may act in a boundless way, with users believing they are protected by freedom of speech in the First Amendment and platforms hiding behind the shield of Section 230 of the Communications Decency Act. This interpretation is often an exaggeration of their rights when it comes to what speech is legally sanctioned, as well as what is socially tolerated. The operating teams of social networks must have a vested interest in limiting negative or controversial speech. As Jhaver et al. explained, “Many participants leave online communities if they become too toxic. Some websites also shut down their comment systems when they are unable to cope with trolls. Therefore, controlling online abuse is critical to maintaining the usability of online spaces” (Jhaver et al. 7).

Midjourney moderators spend more of their time moderating the images generated by users. The relationship between key words and desired images is complicated. The machine learning models could be trained to exclude these key words, but these models lack contextual

knowledge of how different banned words could be used to produce approved visual content. The Midjourney team does not publish the banned words list. The combination of these factors make banned words one of the most discussed topics within the community.

Banned Word List Creation, Testing, and Maintenance

Versions 1-3 of Midjourney were based on the Stable Diffusion text-to-image model. Stable Diffusion is not specific to Midjourney in terms of its usage in online communities. Each version was trained to produce higher quality images, as well as fewer pornographic, gory, or other controversial images. While specific training methods are only known by the Midjourney team, this likely involves filtering the ingested database of text and imagery by a level of controversy. The Midjourney model is then trained to favor generating images that are deemed noncontroversial; by creating the banned words list, the Midjourney team would be able to match language used in prompts with text or images that met that threshold of controversy, and not generate that image. Fractl recalled,

“It used to be we didn’t have to ban many pornographic terms because anybody who tried to make porn using Midjourney would be disappointed by the results, it would just be like blobs of a person. But as it gets better...we need to be more concerned about this because it’s creating stuff that is like alarmingly realistic.”

The Midjourney team is still developing Version 4 as of November 1, 2022. As this latest version has been developed, other versions have been launched in the community for noise testing. For example, a previous version maintained ratings for how pornographic the image was in a database. Those training the model dismissed that a non-zero percentage of the

images generated were pornographic. However, at the scale that images were being generated in the general community, beta had to be shut down after a day due to how perverted the use of the model quickly became. This testing and releasing is done strictly by the Midjourney team. Moderators and guides provide certain images or users that they deem noticeably explicit for the Midjourney team to test against. So, on the day of that release, moderators and guides were mostly limited to directly dealing with the community rather than the software. Despite this, the moderator and guide community is informed of and shared the current means of moderating the algorithm. They refer to this as the Not Safe For Work algorithm, which applies a numeric value to each banned word. The most inflammatory words have higher values assigned. Both moderators and their bots will be more lenient to words with lower values, such as blood, which could be used for controversial and noncontroversial content (e.g. ‘the blood moon’, ‘vampire drinking blood’, ‘blood orange’, etc.). The highest rated word is the n-word, which results in an immediate timeout, quickly followed by a full banning from the Discord server.

Several actions can be triggered when a user attempts to use a banned word within a prompt. These actions range from automated actions to human interventions. The most common are blocked generations, timeouts, and explanations via direct message. If a user is just beginning to use banned words, they will not receive any direct intervention beyond being unable to produce an image of that banned word prompt (See Figure 9. Timeouts take place when the user receives increasingly longer blocks on their ability to generate any image.

Cody Boy shared a progression of explanations that he may send to one user below. He has multiple other prepared explanations that other moderators and guides can use when they are intervening with users:

(1.) 'A reminder that MJ is asking everyone to keep their pics PG-13. The team does not want to collaborate on images of porn, gore, violence or gross-out material. Don't get yourself banned. OK? Let's not make naked pics of actual people without their consent.' (2.) *'Let's just not make naked pics on MJ at all. OK? Seriously. Don't make porn here. Take a day off to think of stuff you want to make that won't get you kicked out.'* (3.) *'Blocked for a week for making involuntary pornography after multiple warnings.'* (4.) *'No more warnings. Keep it clean or get banned.'*"

As explained within these texts, the enforcement of the banned word list is heavily motivated by the awareness of the Midjourney team and moderator community of how much power undesired content may have over Midjourney's reputation. All moderators mentioned their fear of negative press about Midjourney within their interviews. Cody Boy specifically is thankful that, "there's been a lot of news about AI art with people making all of this wonderful stuff and...they are threatening artists, as opposed to threatening morality." The moderators apply these guidelines differently to trial users compared to subscribed users. Moderators are likely to ban a trial user that uses their limited, free prompts on controversial content. Once users start paying, giving back to the community, or start heavily using the system, moderators have more criteria evaluate beyond just the banned words they chose to try.

These guidelines are difficult to enforce consistently at the scale needed for a massive online community, so the moderator team uses Discord bots. These bots live within the

moderator and guide channels; most moderators spend ample time reviewing the banned bot logs, which show what users attempted to generate. There are also commands to make it simple to review the demographic information and history of these users, such as if they pay a subscription, the number of warnings they have received, and an easily accessible link to their personal galleries to review and hide images. Moderators admitted that they must rely on both the algorithms and user reports in order to handle the volume of content. Of course, even if the majority of the community abides by these behavioral rules, there are nonobvious exceptions that make judgment murkier. Fractl confessed, “Our goal is to punish people who are trying to create really bad images using Midjourney, but I think we also end up punishing people who are just curious and are testing what the limits are.”

Feedback and Impact of Banned Word List

It is important to note that the controversy of any single word is subjectively defined. Certain banned words can create a back-and-forth between the users and moderators where a user continues trying iterations of a banned word, including misspelled, translated, and synonymous versions of the word, and the moderator must continuously update the banned word list. This greatly informs the Midjourney team’s decision to not publish the banned word list. Publishing the list is a common request in office hours, and most moderators have received this request directly. Fractl defended the banned word list stating, “This isn’t like Photoshop, it’s not a private tool that you’re using on your own for your own use. It’s a community space and everything that you make is visible, and that definitely changes what is considered to be acceptable.” The Midjourney team seems to be aware of the ‘broken windows’ theory in social

psychology, which is a phenomenon that explains why civil disorder, crime, and anti-social behavior increase as a community is more exposed to it.

Regardless of how commonly an average user sees undesired content, it is a daily occurrence for the Midjourney team and its moderator and guide community. Most moderators likely have not maintained this role long enough to experience long-term mental health effects from the consistent exposure to undesirable content and behavior. However, moderators have already been tasked with numerous sociopolitical debates due to the banned word list. Fractl shared two examples,

“So the term gay: we know that word because it’s included in so many kink searches, we know what it creates. But we’re not going to ban that word because it’s an identity, like we would not want to make that statement. We didn’t ban the phrase ‘sex’ so words that contain the word sex within them, that would ban the words homosexual, intersex, transsexual, and we don’t want to be doing that.”

While these moderators volunteered to make the Discord server a better place, they were not trained on how to handle such social issues. As Cody Boy lamented, "The (moderators and guides) are working really hard to make something beautiful, and when they see people using their work and redirecting it to make something horrible, it's discouraging." This has made moderators and guides spend less of their time creating art or enjoying the community, and more time dealing with inflammatory text and imagery. Even though this community likely knew the ramifications of what they volunteered for, the proportion of moderators and guides to controversial content is harrowing.

Getting Started with Midjourney: First impressions of the Community

The onboarding process to Midjourney is minimal and hands-off. Upon joining, newcomers receive a message in the “welcome” channel that includes short directions and multiple links. The most informative link is described as “a visual guide on how to get started”, which links out to documentation about Midjourney (See Figure 10).

That URL first takes you to the “Quick Start Guide,” which offers the code of conduct and a basic overview of how to generate an image (“Quick Start Guide September 2022). At the end, the guide encourages the user to go to the “User Manual” or “Resource Links” for more information; the “User Manual” is where the vast technical parameters that prompts can feature are described, and “Resource Links” has 28 external links to user-created guides, as well as a few articles and videos from press (“User Manual”, “Resource Links” October 2022). However, based on observational and interview data, newcomers will not necessarily begin their time with Midjourney reading – they are here to make art after all.

“Chaotic”, “messy”, and “fast” are ways the research team described when they entered the community. This is similar to many of our interview participants. Hematita recalled how she thought about the channels when she first joined. “It was... big. And fast! In the areas dedicated to sharing what we make, it seemed it would get washed away immediately.” In the channels of image creation, where everyone’s images appear with the prompt submitted, one’s own image could be easily lost as there could be 100+ new images created in 10 minutes. It seems members who joined earlier experienced a slow, friendly, and peaceful space. Instead of being overwhelmed, fractl and clarinet shared their feelings of excitement and wonderment. Clarinet

was especially astonished by being able to join this community, “When I joined, there was a feeling of I can't believe I'm here. This is a historic thing. I'm so honored to be here...And it was just everyone was breathless with a sense of blessing.” Clarinet’s “breathless” feeling for Midjourney might be due to the earlier joining time and his familiarity with AI art tools compared to other members who joined later, which Cody Boy and fratcl ratified. Clarinet metaphorically stated, “imagine the Cavemen wandering around and somebody maybe discovered fire and everyone gathers around it.”

Today, publicizing the server and adding a subscription plan changed how members behave now in the community. As clarinet described early members proactively explore by themselves, he described now members “come into the channel and say I need someone to help me create a dog.” He proposed this may be due to the belief that “people believe they've purchased a product and expect customer service”. MariusJuston agreed, “people have preconceived notions about AI where they think they'll have a perfect output, it doesn't, and then they walk away.” We speculate that this preconceived notion may come from the fact that newcomers are attracted to the polished Midjourney work shared externally on social media, which obviously did not show the repeated testing and tweaking of prompt craft.

What's the magic word?

Scholar Raphaël Millière shared, "Prompt engineering is more of an art than a science, especially when it comes to creative uses of AI. It has even been compared to alchemy, or incantation. In addition to having a unique vision for the final products, one must get a feel for the right combination of magic words that will unlock specific styles or subjects with any given

algorithm." (Millière July 17, 2022). Despite offering an accessible entry point to creating high-quality digital art with a few keystrokes and clicks, it is clear that control of the Midjourney interface takes practice, patience, and a healthy amount of luck. How prompts are developed, discovered, shared, and used acts as an important entry point into broader discussions of ownership, moderation, community cohesion, explainability of AI systems, and the future of creative professions.

Control

Whether it is control over the image that is generated, control over the interface, controlling ownership of prompts, or control over the way the AI impacts society, discussions of control permeated interviews with super users and creative professionals across multiple dimensions. These concerns of control represent a greater cultural fear and expectation of AI running amok that cannot be controlled (Chalmers 2022). Concerns of being replaced by machines that cannot be controlled has long lived within the popular imagination and are often discussed in the philosophy channel of the Midjourney Discord server and joked about with Terminator memes across the community. How users approach controlling an algorithmic collaborator that is being updated weekly or ceding control to discover new aesthetic possibilities becomes a means of better understanding the most pressing opportunities and challenges present in the Midjourney community.

Martina compares the UI of Midjourney to the UI of Open AI's DALL-E, favoring the latter: "if they had something more like DALL-E, where you have more control of your prompts and your results in the process. That would be a nicer experience". Framing the Midjourney

interface only in terms of its ability to control contrasts its social features, which allow for the "serendipitous collisions, of just running into other people's stuff". For Oscar, the public nature of Midjourney is a curious affordance when contrasted with other creative tools for generating images, notably Photoshop, "imagine if you're working in Photoshop and you had a setting where it would automatically publish every autosave, every two minutes, auto-save to your personal website. And it would also just like blast out to all your social media, every time you hit save on every project you're working on. That would be an insane feature that no one would ever use. But it's the default from Midjourney." The idea that Midjourney's public affordances enable a unique form of creative community to flourish should come as no surprise, given how often inspiration and motivation are found by users remixing other prompts and styles of prompt craft that appear on their feeds (Manovich 2022). Oscar describes how he found the motivation to progress as a prompt craftsperson in being able to remix the prompts of others, "I think what really compelled me is seeing what other people were making and being inspired by that and then kind of remixing it or taking pieces of it". These public affordances make little sense when seen as a creative tool, but when Midjourney is viewed as a game, they are unmistakable hallmarks to its success.

Several gamified features of the Midjourney Discord Server contribute to creating an engaging user experience. From the ability for a user's creation to make it to the front page of the community tab to unlocking super-user channels where exclusive knowledge is shared, several users mentioned the game-like dimensions of their experience with Midjourney. Ignite compares the "addictive" qualities of Midjourney to being "psychologically addicted to a video game (chasing achievements, high scores, fear of missing out, etc.)". Extending his analysis of

categorizing Midjourney into a genre of games, Ignite describes an exploratory style of interaction where "MidJourney is an endless mode game then. The story is the one we make along the way through our prompts" (Ignite). This value of the endless game mode aligns with David's vision for Midjourney of maintaining a public facing interface. Oscar speculates that this public-facing value could be for building community as much as it could be for building "better for user engagement and driving up metrics and getting more users and making it addicting".

Oscar's usage of Midjourney stands out as being one of the most social of the users that we interviewed, but it is clear that many users appreciate being able to "play" Midjourney as a "multiplayer Imagination game where you're imagining these wild scenarios and saying, what about this? What would this look like? And then someone says, ooh, that's pretty good. What would this look like, and you're able to kind of riff on each other" (Oscar). The public dimensions of the constant flow of /imagine commands streaming across countless channels at ever-growing volumes both frustrate new users and inspire users to become super users, madden moderators, and delight those with imaginations that are just beginning to be unlocked.

Magic

Fictional author Arthur C. Clarke stated, "Any sufficiently advanced technology is indistinguishable from magic." The ideas that "words are magic" and prompt craft is more art than science is common on the Midjourney Discord server. Sometimes breakthroughs in prompting are discovered through a glitch or a "happy mistake" and other times, they are uncovered rigorously through quasi-experimental trial and error. Insights into prompting and

"magic words" are broadly shared throughout the community but can also be kept closely guarded within super-user channels as a form of "arcane knowledge."

In the gulf of explainable AI, users of the Midjourney Discord Server adopt various beliefs to make sense of the system outputs. A lot of chaos goes into what the AI makes, and users will naturally look for patterns to fill in and explain a system that cannot be easily or readily understood. Joseph describes this "superstition and confirmation bias" through the familiar feeling that something "the way this generates images has changed... They must have made some updates to the model based on the images I'm creating, right? Nope. If you flip a coin and you get five heads in a row, the next coin that you flip is not more likely to be tails than any other flip". Even with an understanding of probability and how the machine learning algorithm operates it can be difficult to not imagine there being a "ghost in the machine" or "prompts leaking into one another".

Prompt leakage is a conspiracy rooted in a subchannel of prompt craft. It comes from the understanding Midjourney has a memory and what a user prompts in one instance can affect and enter another future prompt. Despite his technical adeptness at making sense of the Midjourney AI, Joseph describes certain experiences he has had with prompts that made him consider the prompt leak conspiracy noting how, "it's hard to accept the pure science of it that you really are just like, exploring something with a lot of random, chaotic variation". In this sea of random noise, many prompt crafters have tied themselves to the metaphorical mast of several key prompt words like "trending on art station, 4K, Behance, photorealistic, Unreal Engine" in hopes that it will have their prompts sailing to "#award-winning" outputs. Joseph can sometimes use some of these words "to get from 95% to 97%, instead of from like zero to 90,

which I think is what people a lot of times will think that it does". Without a clear means for the interface to explain how images are being affected by different words, many users focus more on outputs than inputs.

Explainability

Knowledge of AI is essential for more sophisticated and nuanced prompt crafting. Many different paths, services, guides, and communities support Midjourney users in explaining the algorithm's inner workings. In the gulf of an easy-to-understand explanation, some users develop conspiracies, superstitions, and metaphors to help craft images of a multidimensional machine learning model that defies the bounds of ordinary human imagination. As much as prompt crafting itself, these different conspiracies, superstitions, and metaphors become valuable currency for improving creative outputs and building diverse subcommunities across the Discord server. In the interviews with different Midjourney users, a diversity of beliefs, opinions, and experiments were used to explain of experience with the Midjourney AI.

Comparing the different AI image generators, it's easy to see clear differences in aesthetic outputs. What many users like about Midjourney images is the "vibe" or artistic sensibility of the algorithm. Shambibble describes the creativity of the Midjourney AI as being uniquely powerful, like "lightning captured in a bottle" compared to other similar AI image generators. Midjourney's ability to "take a vague notion and just like filling in some details that look cool" could be the product of several factors. It might be the product of a "fuzzier model," the community's feedback rating, or the care of its creator, compared to the greater "photorealism" of DALL-E's model which Shambibble described as being a "thicker model" with a

greater amount of data These qualifiers express hard-to-communicate differences between hard-to-understand systems in the emerging textured vocabulary of metaphors and guesses, which we have begun to catalog in this paper's "thick" description (Geertz 2022).

Prompt craft

Prompt craft is the difference between a generated image being professionally commoditized or "looking pretty, but a little off" and collecting dust on a user's hard drive. Some users have clear goals when they imagine with Midjourney while others enjoy the randomness and novelty that come from a machine that can imagine what humans cannot. The words that users prompt to create images with the Midjourney AI open up important questions of how ownership, craft, and creativity are contested, compared, and constructed through diverse systems of belief and aesthetic sensibilities.

Users of Midjourney can seek knowledge to better have their prompts generate images through various channels. One of the most common places to improve skills as a prompt craftsman is in the prompt craft channel. Joseph has been improving the outputs of his prompt craft to enhance his work as a creative director of an indie game studio since he first started as a new user of MidJourney. For Joseph, prompt craft is as much about learning "how to have more control over [the AI]" as it is about helping others learn. Aligned with Oldenberg's model of third places, he considers himself a "regular" in the prompt craft channel (Oldenberg). If the prompt-craft channel is the school, trial and error is the teacher.

Latent Space: Imagining Possibility

What Midjourney is capable of and how it works can be hard to imagine. Super users share various metaphors to make sense of the model set used by the Midjourney AI to render their words into images. Imagining the "latent space" of the AI model as a topography of mountains and valleys was shared by Joseph and Oscar to describe two different features of prompt craft. For Joseph, discovering a "magical prompt" was akin to finding a mountain peak and establishing whether or not it was a local maxima or global maxima for generating a valuable aesthetic output: "when it comes to the magical thinking thing, I kind of think about it in terms of like, peaks, like mountain tops. Where are like, you know, the computer science and word would be low local maxima, versus a global maxima or what have you. That yeah, like, you know, you're you're prompting, you're kind of experimenting with different things. And then you come upon this phrase, and you realize, oh, this phrase is like, at the top of a mountain peak. But you don't know, you can't really see if it's the highest peak around you, or if there's like an even taller one right next to you. So you just kind of have to wander around. And hope that you learn something in the process". For Joseph, wandering the landscape of possibility is important to orient between the relative influence of different words' elevation close to one another for future prompting.

In the same metaphorical mountains of latent space, Oscar turned to the image of a valley when describing bias. Oscar used the metaphor to explain how words in a prompt could be blended adequately: "So if you have a very simple prompt, that's why you're going to be stuck in these sorts of valleys because You don't have enough, other influences. It's like you've been raised in the valley, and you've never left your whole life, you don't know what it's like in the big city of Star Wars or Muppets. So until you, you add those in. Now, it's like your, your

little, your prompt can like travel, you know, between those two points... So it's not, again, it's not a perfect metaphor because it's probably a multi-dimensional space, and not really just like a valley. But you can kind of think of it in that way". It is worth recognizing the nuances of Oscar's understanding of the limitations and usefulness of the metaphor. Paralleling the bias one might expect from someone who "had been raised in the valley" and "never left their whole life" gives a sense of the type of bias one is encountering when attempting to imagine a multi-dimensional space. Oscar built around these metaphorical shortcomings with a more multidimensional model of prompt crafting that used galaxies to show off some of his projects and experiences with Midjourney (See Figure 11).

Clarinet leveraged psychologist Carolyn Dweck's theories of growth and fixed mindset to illustrate the success of certain users over others in Midjourney. Dweck describes in her novel *Mindset: The New Psychology of Success*, people with fixed mindsets as people who believe their intelligence and abilities are stagnant, and that they cannot change or improve the skills they have. People like this tend to enjoy it when they know what is right or wrong, but they have little interest in learning from their mistakes. Conversely, someone with a growth mindset believes that with effort and determination, they can further develop their qualities and skills. They choose to learn from mistakes and look at failures or struggles as opportunities to learn and grow (Dweck 2007). Clarinet stated in his experience, users with a fixed mindset will become aggravated and leave the community. He strongly believed that generating images and varying prompts is the best way to learn Midjourney, rather than reading guides or immediately asking for help. Prompt crafters have developed an ability to act and execute their craft in an integrated manner that cannot be fully captured through an algorithmic list of instructions.

They respond to the idiosyncrasies of AI output to create what they desire and discover the unimaginable. The craftsperson is creative not because of originality or novelty but because of the sensitive, responsive, and well-integrated manner of engaging in sustained and spontaneous activity with the algorithm. There's a tension between letting go of creative control and being rigorous with prompt craft explain why certain people never find their footing in this community.

Community Design Recommendations

The growth that the Midjourney Discord server has experienced is unparalleled. While the research team generated ideas on how the server could be improved, it is necessary to state that the community is operating well for how nascent it is. The structure will likely only remain effective for the near future if the member growth remains exponential. The following recommendations can help retain members who are new, seasoned, and those in power. The recommendations are also prioritized in order of what is most immediately feasible for the community to adopt:

1. Improve Onboarding Process

Personality, timing, and prior knowledge or expectations greatly impact the first impression of the community. However, training is universal, and a clearer, more thorough onboarding process could help standardize user behavior and expectations. Moreover, no one exerts effort into controlling newbie channels, which is rather intimidating because it is users' first exposure to the endless sea of prompts and images without explanation. It is quite likely that new users see a number of different prompt types, prompt lengths, and image styles

within the first minutes within the newbie channels. These channels can confuse users, as there is no clear direction of how to learn from others rather than observing for what the user personally prefers. Multiple interviewees (Spruder, MariusJuston, Hematita) described how much they have learned in the super user clubs as compared to the newbie channels. The educational growth curve is somewhat backwards, and it could be improved if more of the tutorial guides users have created were made obvious and available to newcomers. Currently, newcomers must pay in order to have access to other channels. This limits their access to avenues of more direct or specified education, like the forums, that could pique their interests or refine specific skills. Aditya Johri and Seungwon Yang demonstrated the benefits of scaffolded learning benefitting online community education; specifically, demonstration of the knowledge and marking critical features or concepts were the most used methods that community experts leveraged to educate newcomers (Johri and Yang 1-5). Our data demonstrates there are a variety of users willing to educate, but the access to these experts is not always clear or available to newcomers.

Furthermore, it could be interesting to test how users feel if they are required to test out a few of the most common parameters, like “--stylize” or “--quality”, before being granted access. This would teach users about parameters’ existence, how to refine images, and hopefully, pique their curiosity about other parameters. Conversely, rather than a barrier to more channels, the welcome bot could be programmed to send more advanced welcome messages with this type of information after a user has generated a certain number of images.

2. Increase Usage of Discord Bots

The aforementioned server bots are clearly beneficial to the entire community, even those that are limited to certain user demographics. These bots could have a wider reach, whether that means expanding the functionality of existing bots, developing new bots, or leveraging bots from the Discord community. For example, MEE6 is a highly popular Discord bot used within multiple popular servers like Roblox, Minecraft, and MrBeast Gaming (“MEE6” 2022). This bot can welcome new users with a message and orient them to the server, maintain custom functions, and provide multiple moderation tools that seemingly expand upon the currently existing automoderation toolset. Welcome bots may be able to provide informative disclaimers that could set proper expectations for newcomers. This could also increase the likelihood of newcomers reading the educational documentation that currently is available, but not purposefully put in front of newcomers upon joining. The automoderation bots could also be advanced to rely on less human intervention or approval to timeout or ban users. Once Midjourney becomes more of an official product, the team should dedicate more resources to investigating and building bots that can save newcomers and moderators from the emotional toll of how overwhelming the server can be. Bots could also be leveraged to explore desired features expressed by interviewees and office hours attendees.

3. Increase Places for Socialization

Social gravity is an established social and political theory that illustrates how social ties can help grow networks. The denser the population within a certain network or area is, the more users this area will attract. Midjourney channels are numerous and quite separated, with office hours being the clearest example of an event that strong enough social gravity to build social connections. Users can agree and discuss questions that arise, and they can form

relationships over shared interests and opinions. Unfortunately, this is just a weekly event, which stunts the server's capacity for socialization, and there are many international members missing out due to their respective time zones. This combined with the ability to invite the Midjourney image generation bot into private servers gives room for users to leave the online community. While the user population is not dwindling, this does create the potential for a user retention and user experience problem. Experts can leave (and are currently leaving, in Hematita's case) the community, bringing their knowledge and behavior along with them. The community thrives on educated experts helping newcomers and demonstrating acceptable behavior. Even within the server, the large number of channels severs the user population in separate venues, severing shared learning and bonding in the process. As Clarinet observed,

"They've also talked in office hours about introducing a web-based front-end so you don't have to use the Discord at all, which will draw people away from Discord even more. And I don't know if you're on a website - you're alone in a room with your web page, and you're making art, and you're alone. Even if they added a chat feature, I don't know how people would find each other."

The Discord server could likely be condensed; it may be helpful to longitudinally study which channels are most active or promote the most educational discussions. The Midjourney team should also evaluate how pivotal having the Discord community as the main community is. The ability to generate images outside of the community and the communities across different social media platforms, like Reddit, Facebook, and Twitter, seemingly demonstrate that the Discord server is not meant to be the sole community. The Midjourney team could

better prepare for the potential ramifications of these community design decisions by limiting where Discord servers can congregate and socialize.

4. Address Biases in Database, where Possible

The research team believes that the Midjourney team should prioritize and ritualize bias audits of the Midjourney AI toolset; it is difficult to know how much they are prioritizing this work, due to the black box nature of the tool and due to a lack of public documentation about their efforts to reduce it on the topic. A search of “racial bias” within the server will show images and messages of users reporting racial bias mainly in the “feedback” channel. While the research team agrees that Midjourney is not fully at fault for the biases created in the art due the biases that exist in the Internet at large, to quote American political activist Angela Davis, “In a racist society, it is not enough to be non-racist, we must be anti-racist” (Davis np).

If the Midjourney team can plan to grow their internal testing team, they could dedicate resources to scrubbing and vetting the database to reduce the bias Midjourney has to favor whiteness and masculinity in people, as examples of biases perceived in AI globally. Moderators hopefully would have to address fewer sociopolitical issues while working if the Ai algorithm was less likely to create content that would spark these conversations. Additionally, if images more commonly showcase underrepresented or impaired populations or lesser-known artistic styles, Midjourney users could be inspired and challenged by what they have learned to expect from using generic prompts. This could create an altruistic, educational benefit to the tool that the moderator team seemingly is already trying to informally establish.

Conclusion

There are many exciting possibilities on the horizon for Midjourney. A new algorithm is set to be released any day now, which is reported to have greater coherency and creativity. Promises of animations, videos, and 3D creations lay outside the frame of immediate possibilities. Several participants reported the next leap coming when individual characters can be transformed into a persistent character that allow users to tell stories in comics. The creative tension of the future lies between balancing the desire to control the AI's inputs, outputs, and community and the ability to be pleasantly surprised or even blown away by seeing something unexpected. Despite all the rigor and discipline that goes into prompt craft, AI art generators are still a medium with a mind of its own. Whether one is collaborating with stone and chisel, another human's imagination, or an AI-generated Muppet Space Pirate, it is important to recognize that much of creativity begins where control ends.

Appendix



Figure 1: Midjourney-generated images using the prompt “giant goldfish swimming through forest”

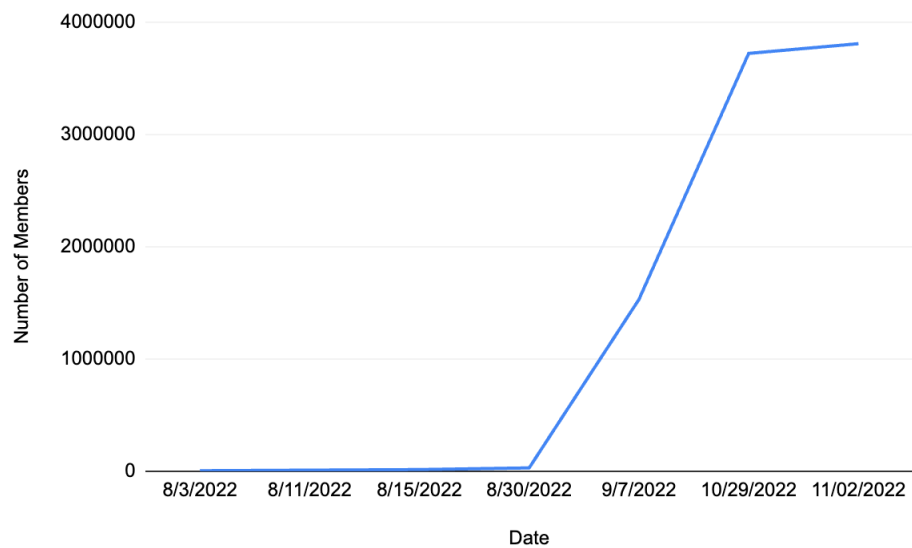


Figure 2: The Growth of the Midjourney Discord server community

MJ Images per User

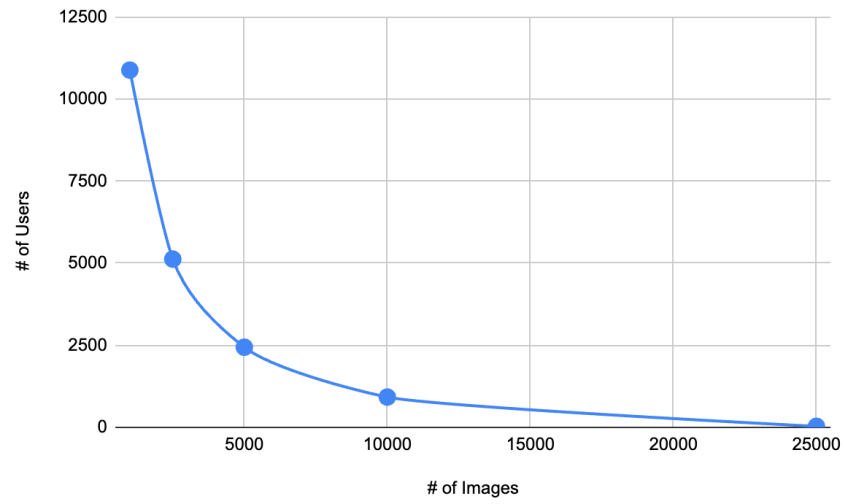


Figure 3: Number of images generated per user



Figure 4: The logo for Midjourney

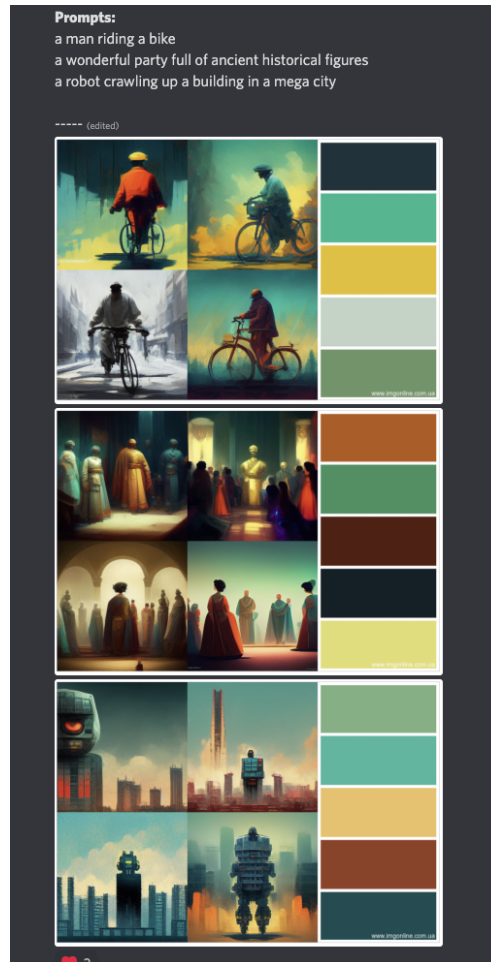


Figure 5: Three examples of the common Midjourney color palette



Figure 6: The member profile of one of the researchers

	Free trial	Basic	Standard	Corporate***
Price	Free	\$10 / month	\$30 / month	\$600 / year
Fast GPU time	25 min* / lifetime	200 min* / month	15 hrs* / month	120 hrs* / year
Relax GPU time	No	No	Unlimited	Unlimited
Metered mode**	No	Yes	Yes	Yes
Personal Bot Chat	No	Yes	Yes	Yes
Private visibility	No	for +\$20 / month	for +\$20 / month	Yes

* 1 generation job roughly takes 1 gpu minute, upscales are longer, variants are quicker
 **Metered mode can be enabled to use additional fast time for \$4 / gpu hour
 ***Corporate plan is required for employees of companies with revenues over \$1Million / year

Figure 7: The member subscription plan options for Midjourney

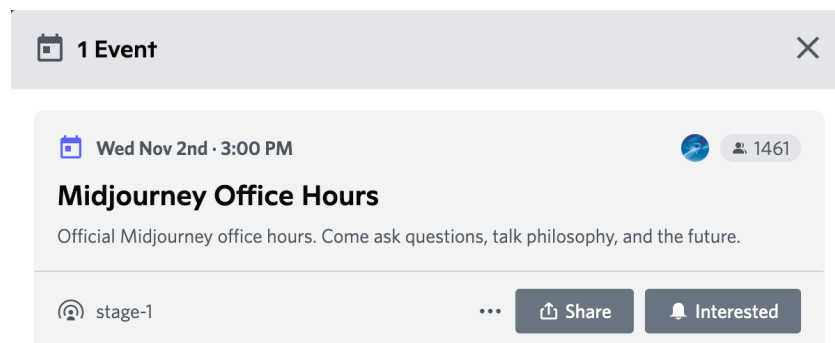


Figure 8: The event reminder for Midjourney Office Hours

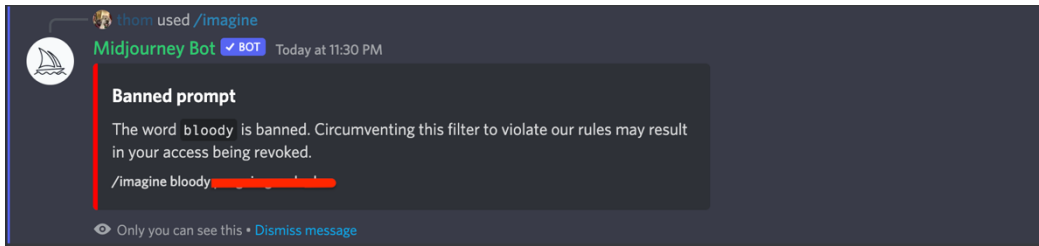


Figure 9: A sample message of the blocked generation of a prompt involving a banned word

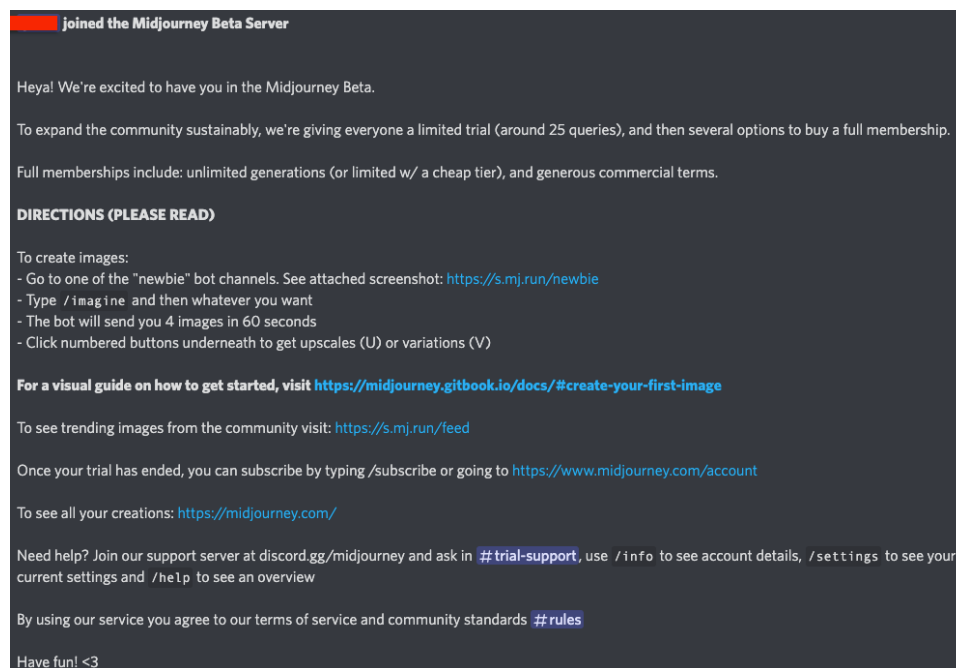


Figure 10: The welcome message every new user receives

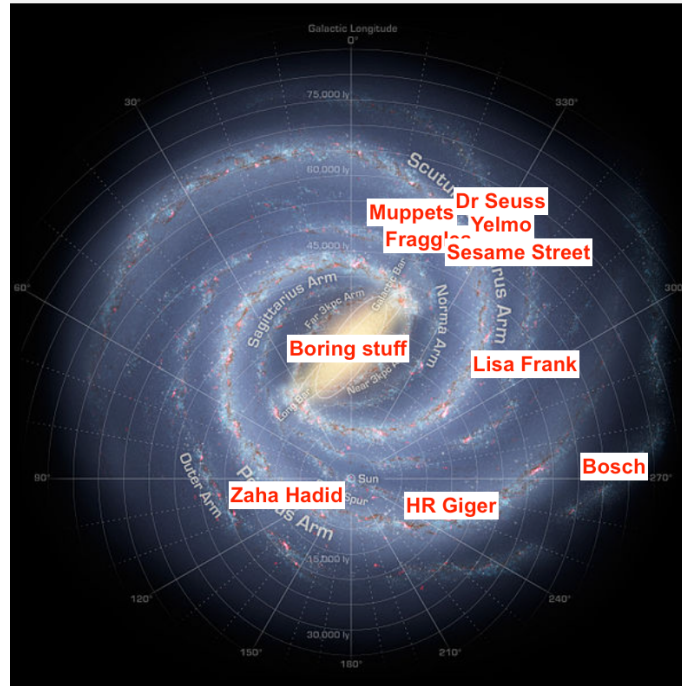


Figure 11: Map of Latent Space of Midjourney AI made by Oscar to represent his experience of the gravitational of the prompts with the strongest orbits in his usage.



Figure 12: Yelmo: Furry Road Collaborative world building project (image by Scott Roberts)

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